

Travel Tales

The Search for Sodom: Claims and Controversy

Part 2

by Lew Toulmin

In early November in this column, I described several candidate locations for the missing sin city of Sodom, all on the west, southwest, or east coasts of the Dead Sea. But there is one more candidate location which has attracted attention – and controversy -- and which has evidence to back up its claim. That is the Tall el-Hammam Excavation Project (TeHEP) located in Jordan, seven miles northeast of the northeast corner of the Dead Sea.

I first got interested in searching for Sodom when I was studying the listings for active digs on the website of the Biblical Archaeological Society at: www.biblicalarchaeology.org/digs/. (This site currently lists about 25 digs in Israel, and one each in Cyprus, Jordan and Turkey. Given the current situation in the Middle East, I would advise sticking to Turkey or Cyprus.)

The TeHEP Sodom effort had launched 16 seasons of winter excavations led by archaeologist Steven Collins from Trinity Southwest University (TSU) of Albuquerque, New Mexico. I was quite keen on researching and finding Sodom, and also liked the fact that the digging would not take place in the broiling heat of the summer, and that each volunteer was not required to stay for months, but rather could participate for one or two weeks.

I was a little nervous to learn that TSU was only accredited by the Southern Association of Colleges and Schools Commission on Colleges (SACSCOC), and not by any US Department of Education recognized associations, and was a faith-based evangelical school apparently intent on proving the literal truth of the Bible. One archaeology professor urged me not to participate in the dig, asserting that the methods used were controversial. But I am not a professional archaeologist, and I wanted to examine the evidence presented, so I signed up.

I recruited a terrific buddy, Kenton Spading, PE, FRGS, with whom I worked on the Amelia Earhart expedition to the Republic of Kiribati, on excavations in Sherwood Forest and at President Madison's Montpelier plantation, and on research into the Glenn Miller disappearance. Arriving at the expedition hotel, we were pleased to find that we would be staying at the impressive and reasonably priced Movenpick Dead Sea Resort, right on the water, rather than in a dusty tent. As a volunteer, I paid for my own airfare, meals and private room in the hotel, about \$100 for optional tours to Petra and Mt. Nebo, and \$775 for the required weekly participation fee. This latter fee was much lower than many archaeological field schools and projects, which can charge over twice as much per week.

The first view of the site was astounding. It was a kilometer (6/10s of a mile) long, 30 meters (about 100 feet) high, and about 200 meters (650 feet) wide, and looked like a giant upside

down canoe. The entire “tel” or “tall” was a man-made mound, made of dozens of layers of human habitation, thousands of tons of crushed building debris and mud bricks, and millions of pottery shards. The tel covered a massive 89 acres. It was first inhabited about 5000 years ago, and last used by the Jordanian military in the 1990s for high-ground, dug-in tank positions in case of an Israeli invasion.

A typical dig day involved a terrific breakfast at the hotel, a bus ride to the edge of the tel, a hike carrying our gear and lunch up to the palace/fortress end of the mound, and hours of careful digging using trowels, brushes and mattocks. The winter weather was a bit cold at first but pleasant as we worked and the desert warmed up. One day the dust-filled wind was so strong that digging was called off. On most days Kenton and I wore masks to keep out the fine dust, although many participants and staff unwisely did not, and some suffered from persistent coughs as a result.

During evening lectures we learned about the evidence for the alleged Sodom site. The theory was in the time of Abraham and Lot, about 1650 BCE in the Middle Bronze Age (MBA), a meteor roared into the region from the southwest, over the Dead Sea. This meteor strike was similar to the “Tunguska event” in Siberia in 1908, in which a stony meteor the size of a 25-story building, traveling over 33,000 mph, exploded about 5 miles above the surface, flattening 80 million trees in a circle over 35 miles in diameter. The blast had 1000 times the energy of the Hiroshima atomic bomb!

The somewhat larger (!) MBA air burst event theoretically blasted Sodom, Gomorrah and the other Biblical “cities of the plain” – which was the circular, well-watered area around the southern mouth of the Jordan River, at the north end of the Dead Sea, not further south. A part of the exploding meteor struck the leading, western tip of the tel, which was a substantial town at the time (not a small village in the south), with about 8000 inhabitants.

None of the residents survived the 3600-degree Fahrenheit blast and the 740-mph shockwave. Clothing, wood and people all burst into flames. The top 40 feet of the palace at the west end of the city was sheared off. Metal and even pottery melted. The intense heat and pressure caused numerous changes in the mineral composition of the affected surfaces. The surrounding region was drenched in salt, so the region was useless for farming and uninhabitable for 600 years. Jericho, only 14 miles to the west, likely had its “walls come tumblin’ down” due to this same terrible calamity.

The evidence for this scenario, published in *Scientific Reports* in 2023, included “melted bits of pottery and mud bricks”, a 5-foot-thick “city-wide” layer of massive destruction in the correct MBA zone, sheared off buildings, an impact calculator, finely fractured sand grains called “shocked quartz” formed under 725,000 psi pressure, tiny “diamonoids” created by immense pressure, “spherules” of iron and sand that melted at over 2900 degrees, “melted platinum, iridium, nickel, gold, silver, chromite and quartz,” and “disarticulation and skeletal

fragmentation” of residents. The mineral findings are reportedly similar to those found at the Tunguska site and the Chicxulub crater -- the meteoric event which killed off the dinosaurs.

Sounds convincing!

But some scientists vigorously disagreed, citing corrections, inconsistencies and vagueness in the assertions and evidence, and stating that the air burst advocates “did not follow well established guidelines and criteria” and that much of the high temperature findings are “well documented in ancient slags” from metal smelting by humans. Critics asked, why were no animal bones and only ten human skeletons found, when there should have been thousands in the destruction layer? Things got nasty when some critics made personal attacks on the meteorite advocates and archaeologists, their degrees, and their small academic religious institutions.

Most scientists are understandably reluctant to make probability assessments of the truth of competing scientific theories. I am not a professional archaeologist or meteor expert, but I am moderately well versed in search and rescue/recovery (SAR) and emergency response. Using the SAR principle of Mattson secret ballot voting for resource allocation, if my team had \$1 million to spend on pursuing the competing Sodom sites, I would vote \$450K for independent verification of the TeHEP site, zero dollars for the Mt. Sodom area on the SW shore of the Dead Sea (due to no archaeological evidence at all), zero for the north center of the Dead Sea underwater, \$450K for the Bab edh-Dhra site on the east central shore of the Dead Sea, and \$100K for “ROW” (rest of world).

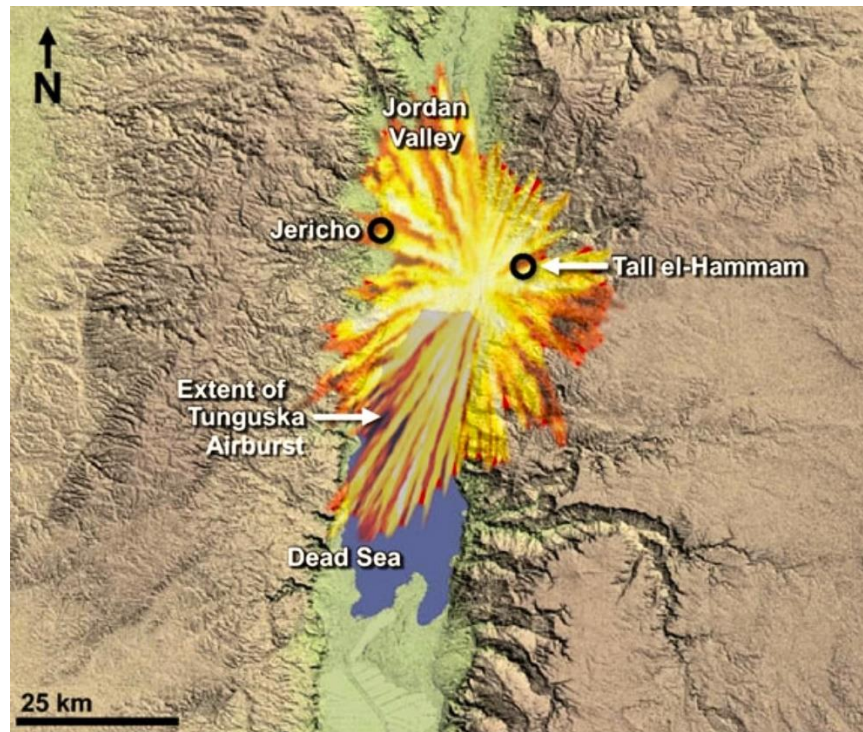
ROW is basically SAR-speak for “we don’t know where the hell it is, but we’ll look around.” I would spend some of that \$100K looking for Gomorrah near TeHEP, to confirm or disprove the meteor theory.

If you want to dig at TeHEP, you may be out of luck. It appears that the digging phase is over, and the team is moving into the multi-year phase of analyzing thousands of boxes of material accumulated over almost two decades of excavation. I hope that this will eventually result in definitive proof – or disproof – of this intriguing and frightening theory of meteoric destruction on a massive, thermonuclear scale.

Meanwhile, keep looking up. All those mystery drones may be aliens scouting for the next place to blast with a meteor. If you see some flashing lights, duck and cover!

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Images and captions:



1. A map of the Dead Sea area, showing Tall el-Hammam overlaid with the estimated Tunguska, Siberia meteoric airburst of June 30, 1908. (Jerusalem (not shown) would be about 30 km (18 miles) WSW of Jericho.) The theory is that a meteor similar to the Siberian one burst over the Dead Sea in the Middle Bronze Age in about 1650 BCE, inundating the region with salt from the Sea, and destroying Sodom, Gomorrah and other “cities of the plain” at the south end of the Jordan River, making the region useless for farming and thus uninhabitable for 600 years. (God was REALLY peeved!)



2. Tall el-Hammam is a “tel” (man-made mound) of millions of pot shards and tons of building debris, in layers, about one kilometer (6/10 of a mile) long and 30 meters (100 feet) high – immense by Middle Bronze Age standards. Note the tiny white patch on the left ridge – that is a person. The view here is from the SE looking WNW towards Jericho. Excavations show that the right high ground was likely a fortress, and on the left was probably a fortified palace.

3. It takes lots of gear and a very sturdy pickup (preferably a Toyota HiLux, not available in the US -- darn!) to run an archaeological dig.



4. The author (right) with Kenton Spading, PE, both Fellows of the Royal Geographical Society and of The Explorers Club, on top of Tall el-Hammam, possibly Sodom. The remaining ramparts of the palace/fort are on the left, and Jericho and Jerusalem are beyond the white distant hills on the right. Jericho, in Israel, is only about 14 miles away. The “cities of the plain” were (in theory) Sodom, Gomorrah and other smaller tels in the fertile green area on the right, watered by the Jordan River and various streams.

5. Don’t step back! A 35-foot-deep hole was required to dig down to the Middle Bronze Age (MBA -- 2100 to 1550 BCE) layer, underneath Iron Age and other layers at Tall el-Hammam. A layer of MBA burning and destruction is in evidence, supporting the theory that this tel is ancient Sodom.

6. Kenton Spading getting his hair done. After each day of digging all volunteers and staff had their hair and clothes blown with a powerful air gun linked to the team’s bus, to remove the many layers of dust and grime.



7. The author holding a large handle to a massive pot, likely from the Iron Age -- thus after 1200 BCE. Still, about 3000 years old, not too shabby.

8. A vitrified pot shard from the Middle Bronze Age layer at Tall el-Hammam. Scientific analysis by the TeHEP team asserts that this and other shards were melted on one side at a very high temperature, about 3600 degrees Fahrenheit, far higher than any man-made pots at the time. This supports the theory that a meteoric airburst created the melting.

9. You can't go near the Dead Sea without getting a Dead Sea mud bath! Very good for the skin -- if you can get it off. Kenton Spading is holding a chunk of Dead Sea salt and a glob of Dead Sea mud.

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